Net force first

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Combine forces before using mass to predict acceleration.

SUCCESS CRITERIA - I can...

1. Find net force before dividing by mass.
2. State acceleration with units and direction.
3. Separate acceleration direction from motion direction.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “net force” and “mass” to explain your answer. Sentence starters are optional.

net force: the vector sum of all forces on one object

mass: a measure of how strongly an object resists acceleration

acceleration: rate of change of velocity

01 Warm-up: bring back an earlier idea

A **4 kg** object hangs still from a vertical string.
1. Use g=**9.8 N/kg**.
2. **Name all forces** on it with sources, sizes and directions.
3. Does object-on-string belong on this diagram?

Sentence starter: In my work, I used … to show … . I labeled it … .

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: ΣF = ma; a = ΣF/m.
Add signed forces on the same object before dividing by mass. N/kg = m/s².

Worked reasoning: A **2 kg** cart has **9 N** right and **3 N** left. Net force=**6 N** right; a=**6/2**=**3 m/s²** right. **Add forces before dividing** by mass.

Your turn: 1. **Draw arrows** for both forces, then a separate net-force arrow.
2. **Explain why 9/2** is not the acceleration.

Sentence starter: For this answer, I treated … as true. I found … . This supports … because … .



2 kg cart: draw the net force separately.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Work from force arrows

A **5 kg** object has **32 N** right and **12 N** left. It currently moves left at **6 m/s**. **Find a** and v after **1 s** under constant forces.

Sentence starter: I found ... [unit]. This value means ... in this situation.



The dot represents the object. Mass: 5 kg; velocity left

Marzano check - circle: 1 2 3

04 Find an unknown force

A **3 kg** cart accelerates left at **2 m/s²**. One force is **4 N** right; the other horizontal force is unknown. **Complete the diagram** and **find the unknown force magnitude**.

Word help: magnitude = size of a quantity without its direction

Sentence starter: I found ... [unit]. This value means ... in this situation.



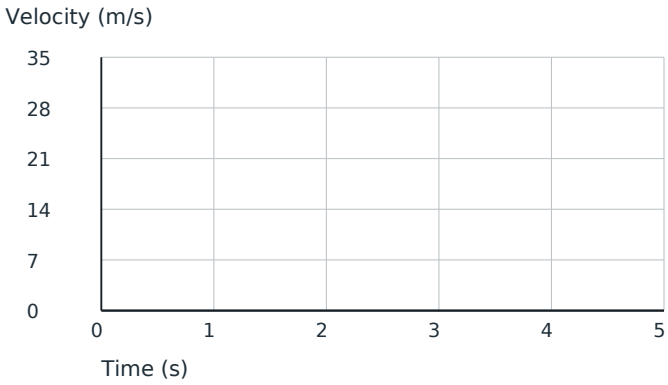
Mass 3 kg; acceleration 2 m/s² left.

Marzano check - circle: 1 2 3

05 From braking to force

A **1500 kg** car brakes steadily from **30 m/s** to rest in **5 s**. **State a positive direction**, plot the velocity-time line, then **find acceleration** and net force.

Sentence starter: I found ... [unit]. This value means ... in this situation.



Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

Positions are **21**, **14**, **7 m** at **0**, **2**, **4 s**. **Predict position** at **9 s**. **Give the rate**, something you are treating as true and a future test.
Separate what was observed from what can be inferred.

Sentence starter: I predict ... . I used ... to make this prediction because ... .

Observation	Supported claim	Missing information
___	___	___
___	___	___

Marzano check - circle: 1 2 3

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Continue down each column.

Sentence starters are optional. You may plan in any language.

07 Exit check

A **4 kg** cart has **7 N** right and **19 N** left.
1. **Find a**.
2. Is it necessarily moving left?
3. **Name the information needed** to find whether it speeds up.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Add and label force arrows.

Draw the given forces; show net force separately.

Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

A **3 kg** cart accelerates left at **2 m/s²**. A rope pulls right with **10 N**.
1. **Find the other horizontal force**.
2. Can the cart be moving right at this instant?

Sentence starter: I found ... [unit]. This value means ... in this situation.

Add and label force arrows.

Label each interaction force on the cart.

Marzano check - circle: 1 2 3